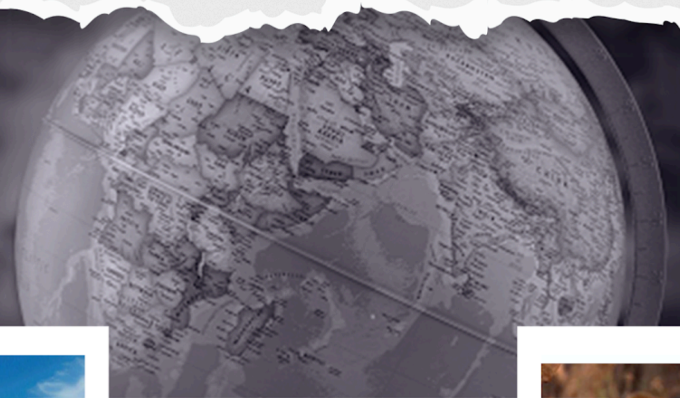
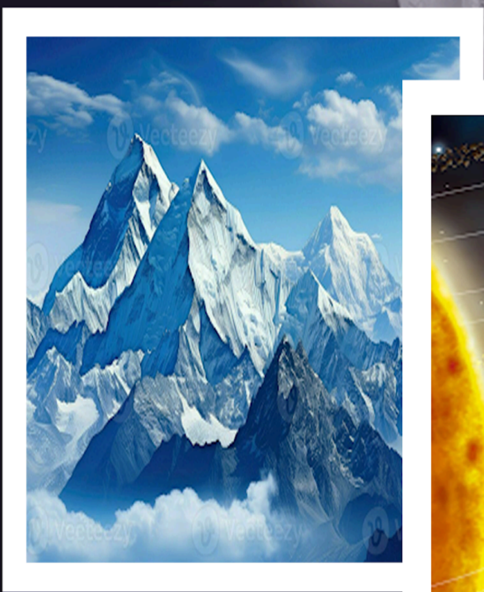
General Knowledge

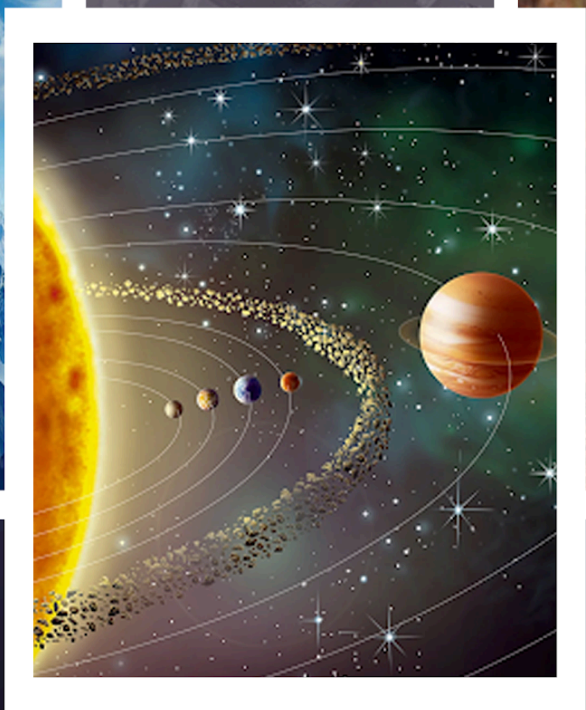
World Geography

STUDY NOTES

Stars and The Solar System







Stars and the Solar System

A solar system is gravitationally bound by the Sun and the objects that orbit around it. Our solar system consists of the **Sun**, and everything bound to it by gravity — the planets **Mercury**, **Venus**, **Earth**, **Mars**, **Jupiter**, **Saturn**, **Uranus**, and **Neptune**, **dwarf planets** such as **Pluto**, dozens of moons and millions of **asteroids**, **comets** and **meteoroids**.

Celestial bodies

The **Universe** encompasses all the **stars**, **planets**, and **galaxies**, and its size remains unknown. Within this vast expanse, there is a spherical region called the **Observable Universe**, which can be observed from Earth. This **Observable Universe** contains various natural physical objects known as **celestial objects**. Examples of **celestial objects** include the **Sun**, the **Moon**, the **stars**, and the **planets** in our **solar system**.

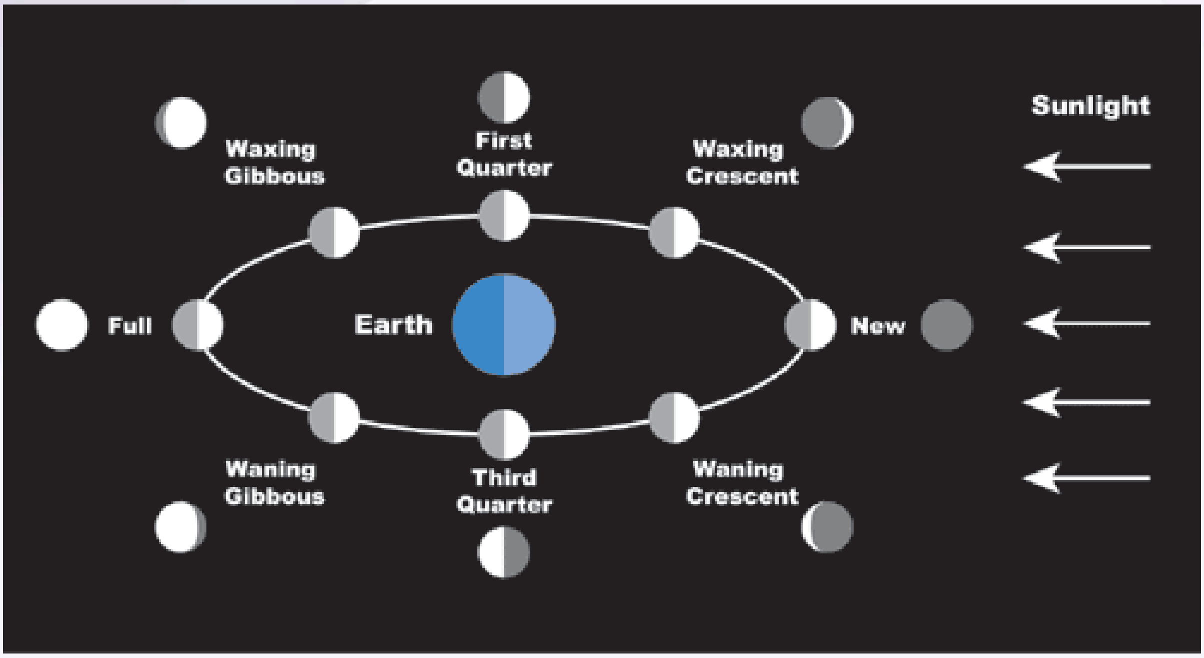
Astronomy

Astronomy is the study of **celestial objects** and the phenomena associated with them and is one of the oldest sciences in the world. It involves studying the **evolution of the universe**, the **motion** of celestial objects, and their **formation**. In ancient times, astronomers observed the sky at night and used these observations to create **calendars** and predict **climate** and **rainfall patterns**. This knowledge helped farmers choose the right **crops** and timing for sowing seeds. It also played a role in determining **festivals** and the occurrence of **seasons** in India. Today, **computers** assist in these **methodical observations** of the sky.

The Moon

The **Moon** is a **natural satellite** of the Earth that revolves around the Earth in a fixed orbit. Similar to Earth, other planets in the **solar system** also have their own moons or natural satellites. The Moon does not emit its own light but **reflects the light** of the **Sun**. As the Moon orbits the Earth, the part facing the Sun is illuminated, while the other part remains dark. This causes the Moon to appear in different shapes, known as the **Phases of the Moon**. The **phases of the Moon** repeat every **29 days**, with **eight major phases**. The day when the entire Moon is visible is called a **Full Moon Day** when sunlight illuminates the whole Moon. As the Moon moves in its orbit, the amount of sunlight it receives decreases, making the Moon appear thinner. On the **15th day** after the Full Moon, the Moon becomes completely invisible, called a **New Moon Day**, as it is positioned behind the Earth. The next day, the Moon appears as a **Crescent Moon**. From there, the Moon grows larger until the next **Full Moon Day**.

- **Waning phase**: The Moon decreases in size.
- **Waxing phase**: The Moon increases in size.



A **Blue Moon** occurs when there are **two full moons** in a single calendar month, with the second one being called the Blue Moon. **The phases of the Moon** are significant in Indian culture and are used to determine the **dates of various festivals**.

The phases of the Moon hold importance in Indian society and culture as different festivals in India are celebrated depending upon the Phases of the Moon.

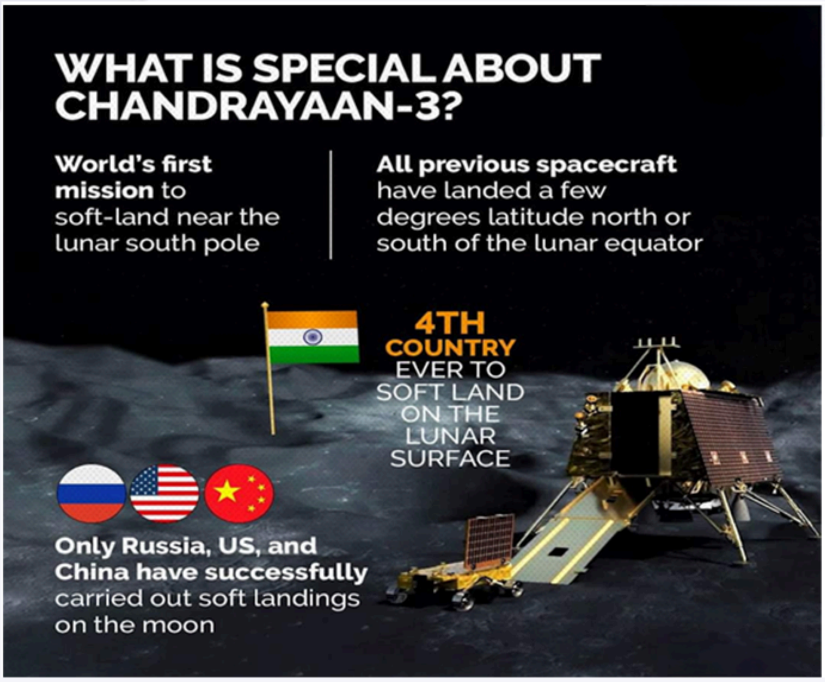
Festival of India	Relation with the Phases of the Moon
Diwali	Celebrated on New Moon Day
Guru Nanak Birthday	Celebrated on Full Moon Day
Maha Shivratri	Celebrated on the 13th night of Waning Moon
Eid-ul-Fitr	Celebrated on the Day following the Crescent Moon

The surface of the Moon

- The surface of the Moon is **barren** and covered with **dust**.
- There are **craters** or bowl-shaped cavities on the Moon's surface.
- The Moon has **steep and high mountains** in huge numbers.
- Some of the mountains on the Moon are as high as the **highest mountains** on Earth.
- There is **no presence of water** or **atmosphere** on the Moon.

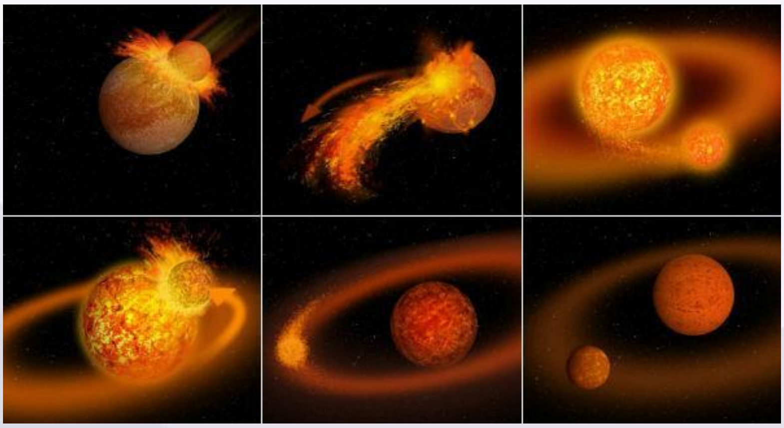
Neil Armstrong was the first person to land on the moon followed by another Astronaut Edwin Aldrin. Neil Armstrong landed on the moon on July 21, 1969.

Chandrayaan-3 is India's third lunar mission and second attempt at achieving a soft landing on the moon's surface. On **July 14, 2023**, Chandrayaan-3 took off from the Satish Dhawan Space Centre in Sriharikota. The spacecraft seamlessly entered lunar orbit on August 5, 2023. The historic moment unfolded when the lander made a successful touchdown near the Lunar South Pole on Aug 23, 2023.



Formation of the moon:

- It is now generally believed that the formation of the moon, as a satellite of the earth, is an outcome of a **'giant impact'** or what is described as **'the big splat'**.
- A body the size of one to three times that of Mars collided with the Earth sometime shortly after the Earth was formed.
- It blasted a large part of the earth into space.
- This portion of blasted material then continued to orbit the earth and eventually formed into the present moon about **4.44 billion years ago**.



Sun

The **Sun** is a **star** at the center of our **solar system** and is the primary source of **light** and **heat** for Earth. It plays a crucial role in sustaining life on our planet and drives many of Earth's natural processes.

Formation of the Sun:

A **nebula** is an interstellar cloud of dust and gases, with a vast size of some hundred light-years in diameter. The **solar nebula** gave birth to our solar system and everything in it. The nebula started its collapse and core formation some 5-5.6 billion years ago and the Sun and the planets were formed about **4.6 billion years ago**.

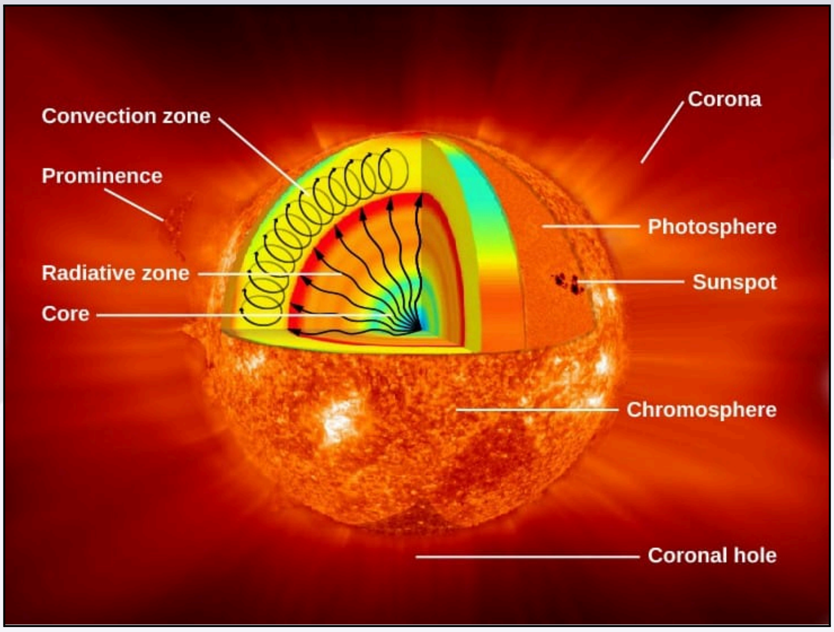
- The nebula began to collapse gravitationally after becoming gravitationally unstable. This probably started because of a nearby **supernova sending shock waves** rippling through space.
- Gravity then caused dust and gas to coalesce at the center of the nebular cloud and as more matter got pulled in, increasing the gravity and pulling even more dust inwards. In the process around 99.9% of the matter was pulled in.
- This resulted in increase in density and temperature and once the center of the cloud became hot enough it triggered **nuclear fusion,** and the Sun was born.

Key Facts About the Sun

1. **Composition**:
    - The Sun is primarily composed of **hydrogen (about 75%)** and **helium (about 24%)**, with trace amounts of other elements like oxygen, carbon, neon, and iron.
    - These elements undergo **nuclear fusion** in the Sun's core, where hydrogen atoms combine to form helium, releasing a massive amount of energy in the process.
2. **Size**:
    - The Sun is **vast** in size, with a **diameter** of about **1.4 million kilometers** (870,000 miles), which is roughly **109 times** the diameter of Earth.
    - Its **mass** accounts for about **99.86%** of the total mass of the entire solar system.
3. **Energy and Light**:

- The Sun is a **powerhouse** of energy, producing energy through **nuclear fusion** in its core. This process converts hydrogen into helium, releasing an enormous amount of energy in the form of **light** and **heat**.
- The Sun's **energy** is responsible for **photosynthesis** in plants, the **water cycle**, and weather patterns on Earth.

4. **Structure of the Sun**:
    - The Sun consists of several layers:
        - **Core**: The innermost part, where **nuclear fusion** takes place, producing energy.
        - **Radiative Zone**: The energy produced in the core moves outward through this layer by radiation.
        - **Convective Zone**: Here, energy is transferred through convection, where hot gases rise and cooler gases sink.
        - **Photosphere**: The Sun's **visible surface**, from which light is emitted.
        - **Chromosphere**: The layer above the photosphere, visible as a red ring during a **solar eclipse**.
        - **Corona**: The outermost layer, visible during a **total solar eclipse**, which appears as a halo of light.
        - **Sunspots** are dark patches on the Sun's surface that appear cooler than the surrounding area, with temperatures ranging from 500 to 1500°C lower. These spots have a lifespan that varies from a few days to several months. It is believed that the Sun becomes about 1% cooler during sunspot activity, which may influence Earth's climate by slightly altering solar radiation levels.
        - **Solar wind** is a continuous stream of energized, charged particles, primarily **electrons** and **protons**, that flow outward from the Sun at speeds up to 900 km/s. The temperature of the solar wind is about 1 million°C, and it plays a key role in interacting with Earth's magnetic field and atmosphere.
        - **Solar flares** are intense bursts of energy and light produced by magnetic anomalies on the Sun's surface. These flares appear as bright spots and are often associated with eruptions of gas. As solar flares move through the Sun's outer atmosphere, the temperature of the **corona** rises to between 10 and 20 million °C.
        - **Solar prominences** are large arcs of gas that extend from the Sun's surface, held in place by the Sun's magnetic fields. These loops of gas can reach hundreds of thousands of miles into space. Eventually, many prominences will erupt, releasing vast amounts of solar material into space.



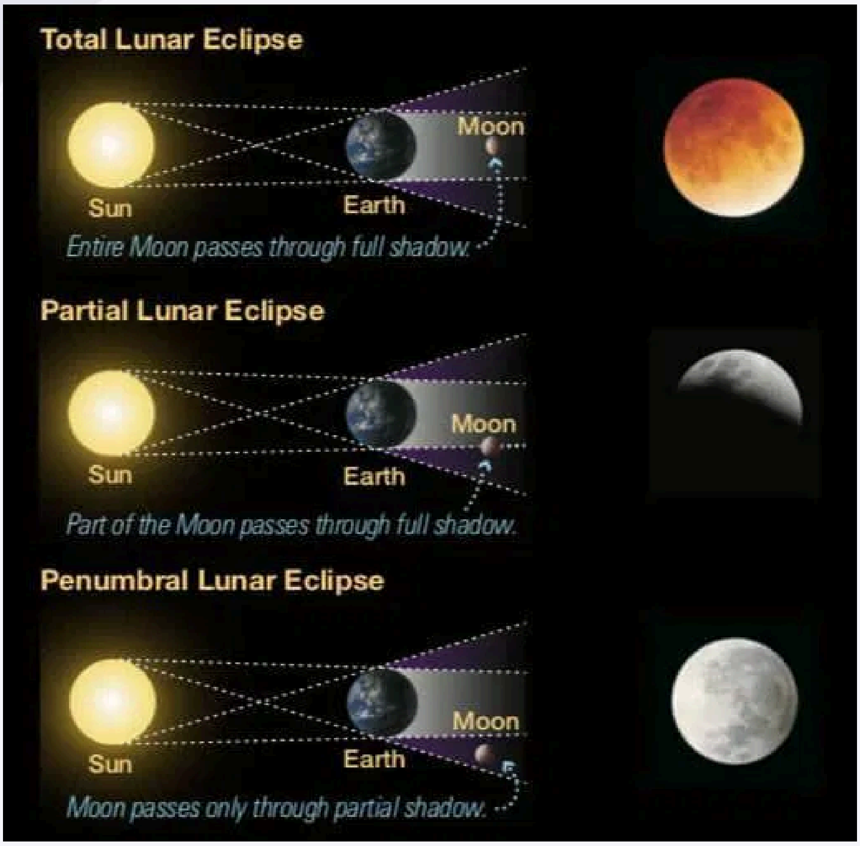
5. **Sun's Lifespan**:
    - The Sun is currently about **5 billion years old** and is considered to be in the **middle** of its life cycle.
    - It is estimated that the Sun will continue to shine for about **5 billion more years** before it exhausts its hydrogen fuel.
    - At the end of its life, the Sun will **expand** into a **red giant**, possibly engulfing the inner planets (Mercury, Venus, and Earth), and eventually shed its outer layers, leaving behind a **white dwarf**.
6. **Sunlight**:
    - The light from the Sun travels to Earth in about **8 minutes and 20 seconds**.
    - The Sun's energy is essential for life on Earth, affecting everything from **climate** to **weather** patterns.
7. **The Sun's Influence on Earth**:
    - The Sun's gravity holds the **solar system** together, keeping planets, moons, asteroids, and comets in orbit.
    - It influences **Earth's climate**, creating seasonal changes, weather, and ocean currents.

Lunar and Solar Eclipse

Lunar Eclipse

- A **lunar eclipse** occurs when the **Earth** comes between the **Sun** and the **Moon**.

- During this event, the Earth’s shadow falls on the Moon, causing it to darken or take on a reddish hue.
- A lunar eclipse can only happen during a **Full Moon**.
- It is visible from anywhere on the night side of the Earth.
- There are three types of lunar eclipses: **total lunar eclipse**, **partial lunar eclipse**, and **penumbral lunar eclipse**.



Total Lunar Eclipse

- The entire Moon passes through the Earth's **umbra**, resulting in a fully darkened Moon.
- The Moon may take on a reddish color, often called a **"blood moon"** due to the Earth's atmosphere scattering sunlight.

Partial Lunar Eclipse

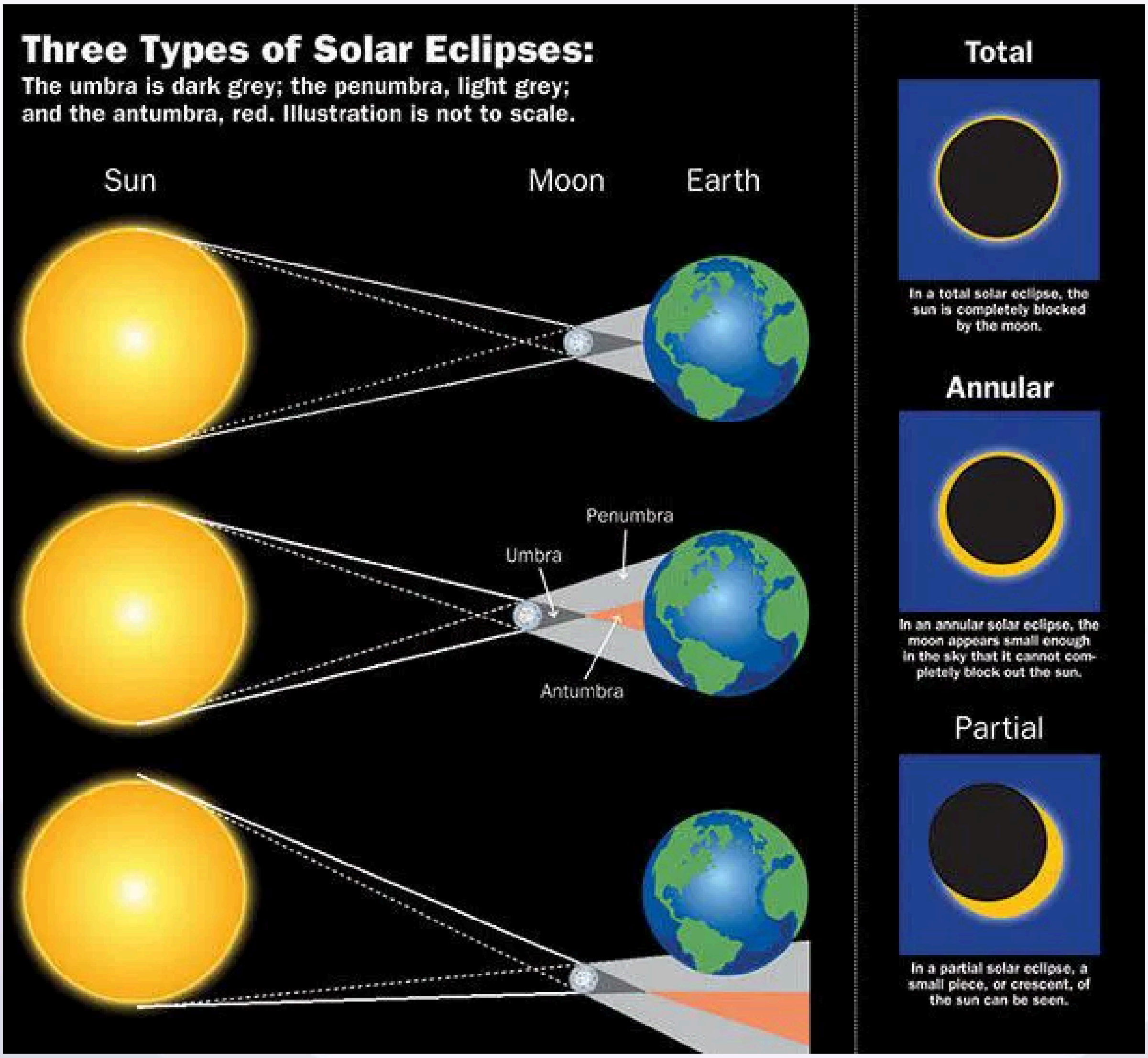
- Only a portion of the Moon passes through the Earth's **umbra**, causing only part of the Moon to darken.

Penumbral Lunar Eclipse

* The Moon passes through the Earth's **penumbra** (outer shadow), causing a subtle darkening, often hard to observe.

Solar Eclipse

* A **solar eclipse** occurs when the **Moon** comes between the **Earth** and the **Sun**.
* The Moon casts a shadow on the Earth, causing the Sun to appear partially or completely obscured.
* A solar eclipse can only happen during a **New Moon**.
* Solar eclipses are only visible from specific areas on Earth, where the shadow of the Moon falls.
* There are three types of solar eclipses: **total solar eclipse**, **partial solar eclipse**, and **annular solar eclipse**.



Total Solar Eclipse

- The **Moon** completely covers the **Sun** from the perspective of an observer on Earth, casting a **shadow** on the Earth.
- The **corona** (outer atmosphere of the Sun) becomes visible.

Partial Solar Eclipse

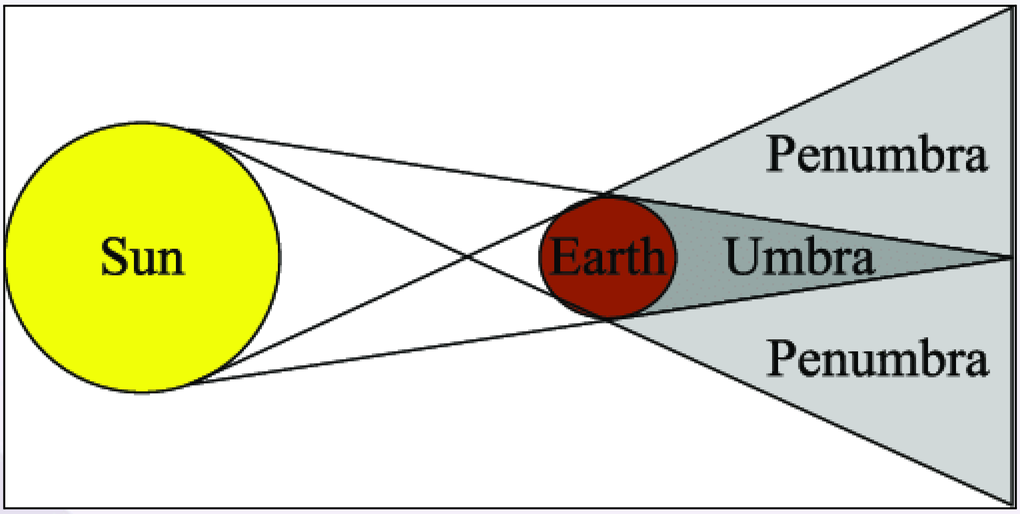
- The **Moon** covers only part of the **Sun** from the Earth's perspective, leaving the rest visible.

Annular Solar Eclipse

- The **Moon** is farther from the Earth, so it appears smaller than the **Sun**, resulting in a ring of sunlight around the Moon, known as the **"ring of fire."**

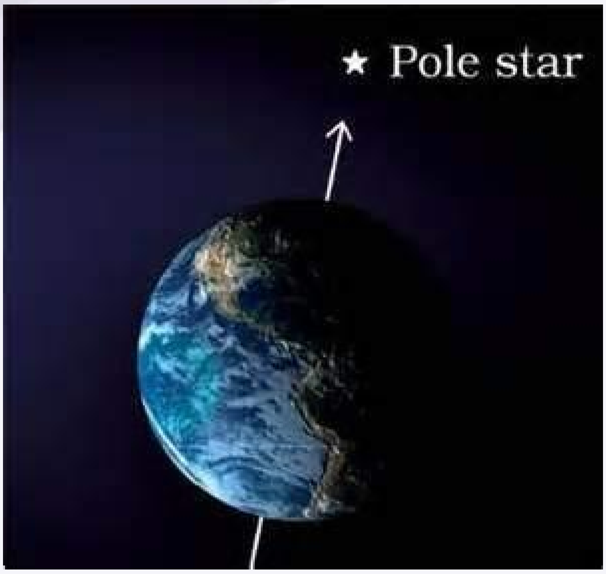
Associated Terms

- **Umbra**: The dark, central shadow of the Earth or Moon during an eclipse.
- **Penumbra**: The lighter, outer shadow that causes partial obscuration.
- **Antumbra**: The area where the Moon is too far from the Earth to completely cover the Sun, seen during an annular eclipse.
- **Saros Cycle**: A period of approximately 18 years, 11 days, and 8 hours after which eclipses repeat in a nearly identical way.



The Significance of the Pole Star (Polaris)

- The **Pole Star** (Polaris) is located near the Earth's **axis of rotation**, making its position **fixed** in the sky.
- Unlike other stars, its position remains **constant** as Earth rotates.

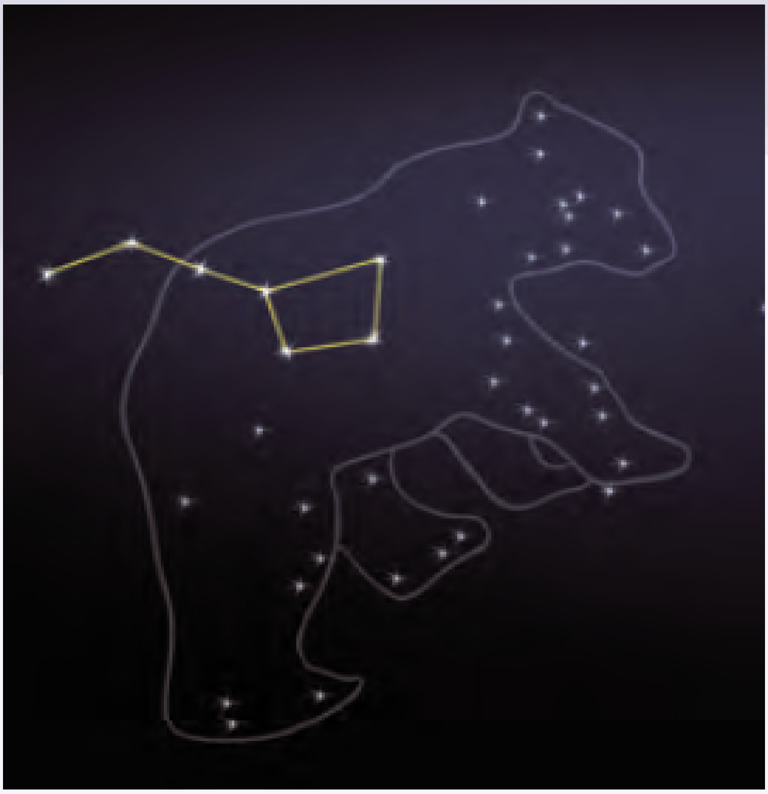


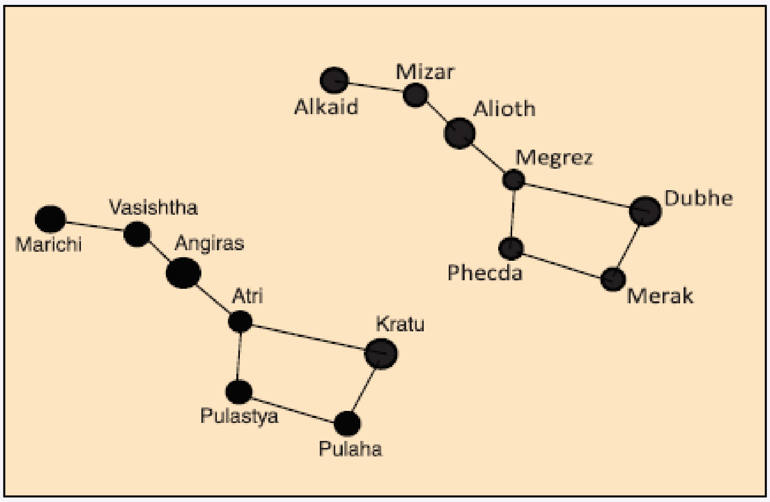
Constellations

- A **constellation** is a group of stars forming a recognizable pattern or shape.
- Ancient people used **constellations** for **navigation** and to identify **stars**.
- Constellations are made up of **several stars**, but only the **brightest stars** are visible from Earth.
- Stars in a constellation may appear to be close, but they are often at **vastly different distances**.

Major Constellations

1. **Ursa Major (The Great Bear / Saptarishi / The Big Dipper)**
    - Comprises **seven major stars** forming the shape of a **ladle** or **question mark**.
    - The **handle** is formed by **3 stars** in a row, and the **bowl** is made of **4 stars**.
    - In **Indian mythology**, these seven stars represent **seven saints** (Saptarishi), who preserve the knowledge of the **Vedas**.





2. **Locating the Pole Star using Ursa Major**:
    - The stars **Merak** and **Dubhe** form the **pointer stars**.
    - An imaginary line between them, extended, leads to the **Pole Star**.
3. **Orion (The Hunter)**
    - One of the most prominent constellations, containing **7 or 8 bright stars**.
    - The **belt** consists of **3 stars** in a line, with **4 other stars** forming a quadrilateral.
4. **Sirius Star**:
    - The **brightest star** in the sky is located in the **Orion constellation**.
    - An imaginary line through Orion's belt leads to **Sirius**.
5. **Cassiopeia**

- Located in the **Northern Hemisphere**, visible during the **winter** months.
- The shape resembles a **W** or a distorted **M**.
- Named after **Queen Cassiopeia** from **Greek mythology**.

6. **Leo Major**
   - Contains several **bright stars** and can be seen from both the **Northern** and **Southern Hemispheres**.
   - The constellation appears **upside down** in the Southern Hemisphere.
   - Recognized **6000 years ago**, it is linked to the **Adventures of Hercules** in mythology.

Stars

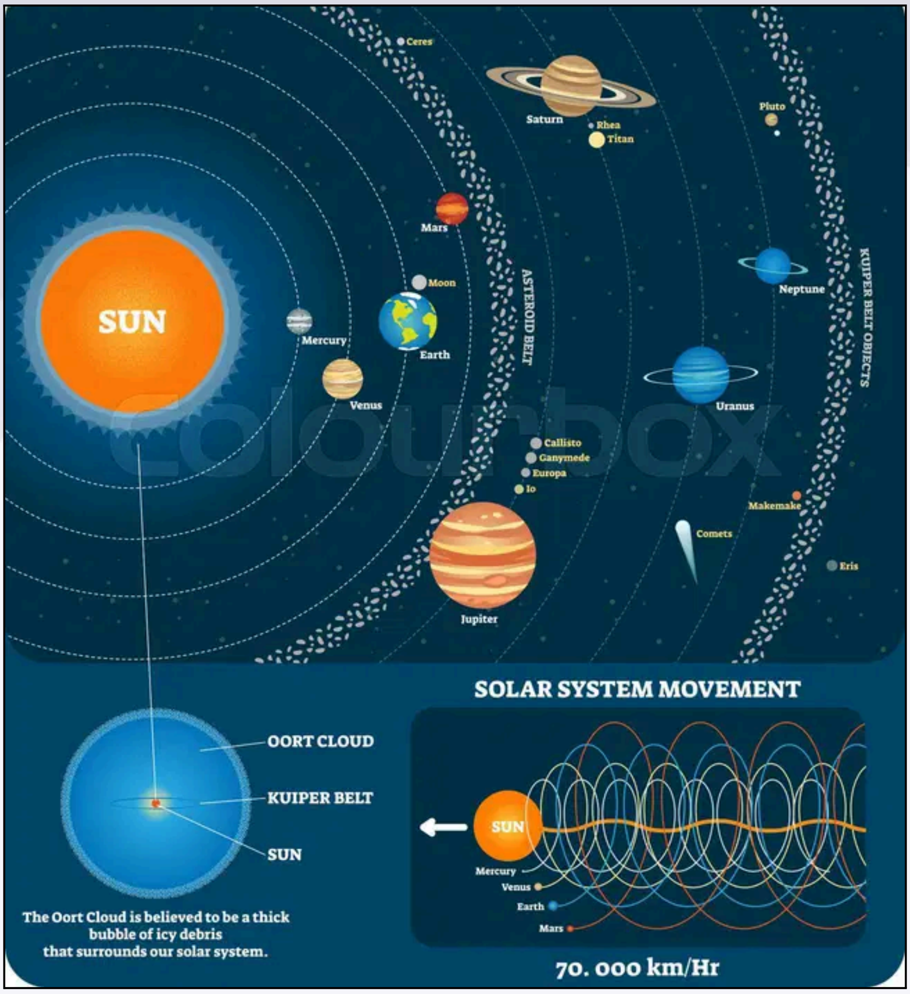
- The **stars** are celestial bodies made up of **hot gases**, mainly **helium** and **hydrogen**.
- All stars **emit their own light**.
- Stars **differ** in **brightness**, **size**, **colors**, and **temperature**.
- Despite being **massive**, stars appear as **point-sized objects** from the Earth's surface due to their vast distance from the Earth.
- Stars have a **life period**, taking millions of years to be born, live for tens of millions of years, and eventually end.
- Because of their long **lifespan**, changes in stars occur gradually, making them appear **permanent** to us.

The Planets:

A **celestial body** moving in an elliptical orbit around a star is known as a **planet**.

An Astronomical Unit (AU) is the average distance between Earth and the Sun, which is about **150 million km**.

Kuiper Belt – It is similar to the asteroid belt but far larger, that extends from the orbit of Neptune at 30AU to 50AU. It may be treated as the extent of the Solar System, as here the gravitational pull of the Sun matches that of several other stars.



Formation of Planets:

- After the formation of the Sun, only **0.1%** of matter remained in the orbit around the Sun, causing the randomly shaped gas cloud to form a flat disc shape, called the **protoplanetary disc**, which was where the planets formed.
- Within the solar nebula, the dust particles in the gas occasionally collided and clumped together. This resulted in **accretion**, the microscopic particles formed larger bodies that eventually became the infant stage of a planet with sizes up to a few kilometers across.
- The infant planets grew in size through accretion to form **proto-planets**. Gradually they got larger and larger, sweeping up all the leftover dust, and other proto-planets until they grew into the **planets**.
- Several rocks that escaped the pull of planets were left as **asteroids**, scattered through the solar system. Many of these rocks orbit the Sun in an area **between Mars and Jupiter** known as the asteroid belt.

Iron Catastrophe and Planetary Differentiation -

- Earth was formed about 4.5 billion years ago; during that time, it was a uniform ball of hot rock. Due to radioactive decay and leftover heat from planetary formation, the ball got even hotter. And, after about 500 million years, our young planet's temperature heated to the melting point of iron — about 1,538° Celsius. This pivotal moment in Earth's history is called the **Iron Catastrophe**.
- The iron catastrophe allowed greater, more rapid movement of Earth's molten, rocky material. The buoyant material, such as silicates, water, and even air, stayed close to the planet's exterior and droplets of iron, nickel, and other heavy metals gravitated to the center of Earth, becoming the early core. This important process is called **planetary differentiation**.

Planet Name (English)	**Surface Temperature (°C)**	**Period of Rotation**	**Period of Revolution**	**Distance from Sun (AU)**	**Diameter (km)**	**Moons**	**Density (gm/cm³)**	**Rotation (East to West/West to East)**
Mercury (Budh)	430 (day) / -180 (night)	58.6 Earth days	88 Earth days	0.39	4,880	0	5.427	West to East
Venus (Shukra)	465	243 Earth days	225 Earth days	0.72	12,104	0	5.243	East to West
Earth (Prithvi)	15	24 hours	365.25 Earth days	1	12,742	1	5.514	West to East
Mars (Mangal)	-60	24.6 hours	687 Earth days	1.52	6,779	2	3.933	West to East
Jupiter (Brihaspati)	-110	9.9 hours	11.86 Earth years	5.2	139,820	95	1.326	West to East
Saturn (Shani)	-140	10.7 hours	29.46 Earth years	9.58	116,460	82	0.687	West to East

Uranus (Arun)	-195	17.2 hours	84.01 Earth years	19.18	50,724	27	1.27	East to West
Neptune (Varun)	-200	16.1 hours	164.8 Earth years	30.07	49,244	14	1.638	West to East

Mercury

- It has almost no atmosphere to retain heat, it has surface temperatures of −173 °C at night to 427 °C during the day, and they vary more diurnally than on any other planet in the Solar System.
- Mercury is smaller than the largest natural satellites in the Solar System i.e. **Ganymede (moon of Jupiter)** and **Titan (moon of Saturn)**. However, Mercury has more mass than Ganymede and Titan.

Venus

- Venus is the brightest planet in the solar system and is the third brightest object visible from Earth after the sun and the moon, it has the highest **albedo** due to the highly reflective sulphuric acid that covers its atmosphere.
- Venus is sometimes called **Earth's twin** because of their similar size, mass, proximity to the Sun, bulk composition, and presence of similar physical features such as high plateaus, folded mountain belts, numerous volcanoes, etc.
- The surface of Venus is totally **obscured** by a thick atmosphere composed of about **96% carbon dioxide**, covered with clouds of highly reflective sulphuric acid.
- It has the **densest atmosphere of the four terrestrial planets**. The atmospheric pressure at the planet's surface is 92 times that of Earth.
- Venus is by far the **hottest planet** in the Solar System, even though Mercury is closer to the Sun. This is because of the **greenhouse effect** arising from high concentrations of CO_2 and a thick atmosphere.
- A day on Venus is equivalent to 243 Earth days and **lasts longer than its year** (224 days). It also rotates in the **opposite direction i.e. clockwise** to most other planets.

Mars

- Mars is often referred to as the “Red Planet” because of the **reddish iron oxide** prevalent on its surface.
- Mars has a thin atmosphere and has surface features ranging from impact craters of the Moon to the valleys, deserts, and polar ice caps of Earth.
- There is no liquid water on the surface of Mars due to low atmospheric pressure which is smaller than 1% of the Earth’s.
- Landforms visible on Mars strongly suggest that **liquid water has existed** on the planet’s surface.
- Mars lost its magnetosphere 4 billion years ago, possibly because of numerous asteroid strikes, so the solar wind interacts directly with the Martian ionosphere, lowering the atmospheric density.
- The atmosphere of Mars consists of about **96% carbon dioxide**, 1.93% argon, and 1.89% nitrogen along with traces of oxygen and water. **Methane** has been detected in the atmosphere which **may indicate the existence of life**.

Jupiter

- It is mostly gas and liquid swirling in complex patterns with **no solid surface**.

- Jupiter's four large moons are called the **Galilean satellites** because Galileo discovered them. They are - **Io, Europa, Ganymede, and Calisto**.
- **Ganymede** is the largest natural satellite (5,268 km in diameter) in this solar system **and is larger than Mercury, and three times larger than the earth's Moon** (3,474 km in diameter, the **fifth** largest moon).
- Because of its **rapid rotation** (once every 10 hours), the planet's shape is that of an oblate spheroid (a slight bulge at the equator). The outer atmosphere is visibly segregated into several bands, resulting in turbulence and storms.

Saturn

- The most distinctive feature of the planet is **Saturn's rings** which are probably made up of billions of particles of ice and ice-covered rocks.
- **Titan** is the second-largest moon in the Solar System and it is the only satellite in the Solar System with a substantial atmosphere mostly made up of nitrogen.

Uranus

- In contrast to all other planets, it is **tipped and spins on its sides**, that is its axis of rotation lies in nearly the plane of its orbit. Therefore, the poles of Uranus lie in a plane where equators of other planets lie.
- **Venus** and **Uranus** have a strange **clockwise** rotation, i.e., opposite of Earth's rotation.

Neptune

- Uranus and Neptune (the ice giants) are called the twins of the outer solar system.
- They are surrounded by a thick atmosphere of hydrogen and helium and contain a higher proportion of "ices" such as water, ammonia, and methane ice giants" to emphasize this distinction.
- Neptune has the strongest sustained winds (2,100 km/h) of any planet in the Solar System.

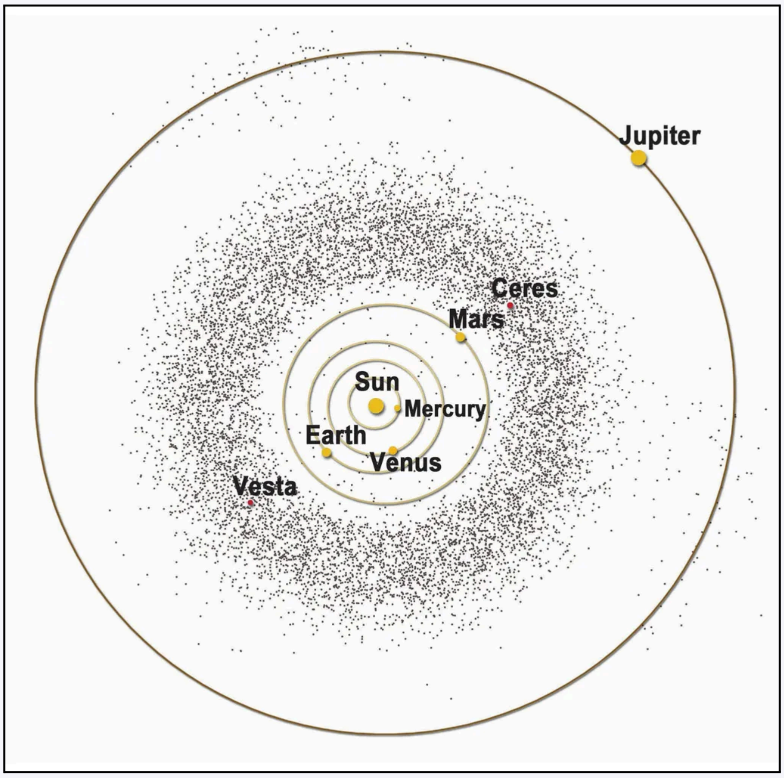
Asteroid Belt

It is a remnant of planetary formation, that failed to coalesce because of the gravitational interference of Jupiter and which circles the Sun in a zone lying between **Mars** and **Jupiter**.

- Asteroids are mainly made up of refractory rocky and metallic minerals.
- Asteroids range in size from hundreds of kilometers across to microscopic.
- All asteroids except the largest, **Ceres**, are classified as small Solar System bodies.

Ceres

- Ceres is the **largest asteroid** (946 km in diameter), a proto-planet, and a **dwarf planet**.
- Ceres has a mass large enough for its own gravity to pull it into a spherical shape.



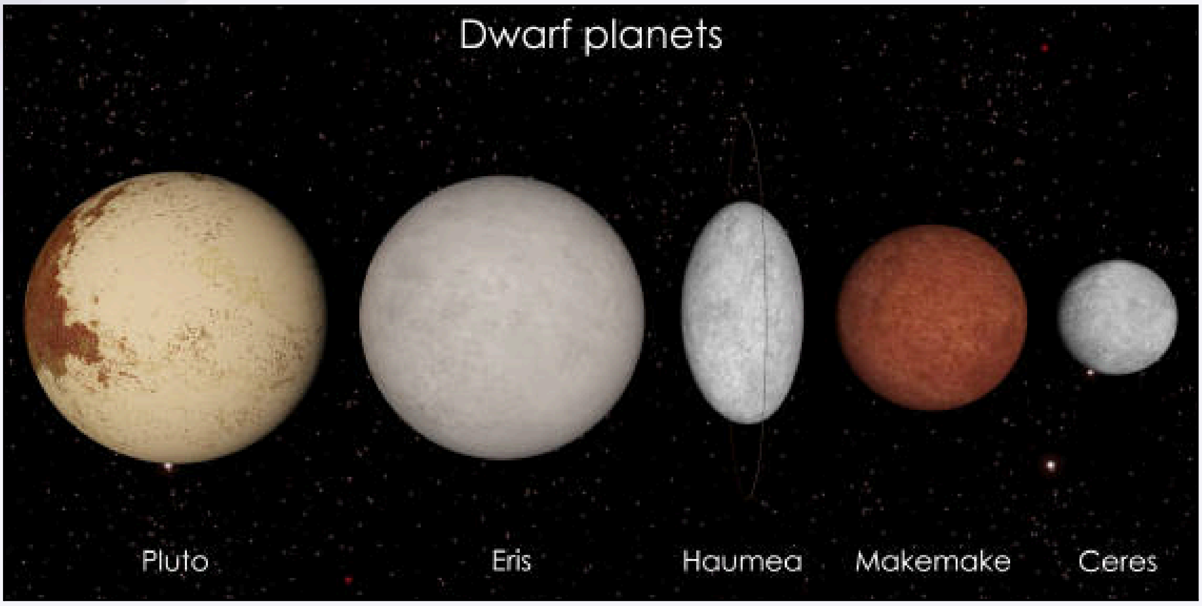
International Astronomical Union’s definition of a Planet -

A Planet is an object that:

- Orbits the sun
- It has sufficient mass to assume **hydrostatic equilibrium** — a nearly round shape
- It is not a satellite of another object
- It has removed debris and small objects from the area around its orbit

IAU’s definition of a Dwarf planet -

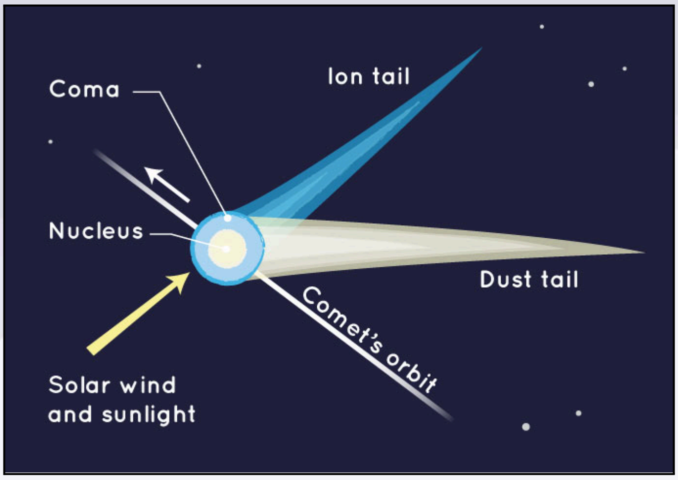
- A **dwarf planet** is an object that meets planetary criteria except that it has not cleared debris from its orbital neighborhood.
- **Pluto** is a part of the **Kuiper belt** that contains millions of rocky and icy objects. Also, there are numerous other objects in the Kuiper belt that are of similar size to Pluto. E.g. Eris (diameter: 2,326 km).
- **Hence,** Pluto (diameter: 2,377 km) (Kuiper belt) **was voted by IAU as a dwarf planet just like** Ceres (asteroid belt) **and** Eris (diameter: 2,326 km) (Kuiper belt)**.**



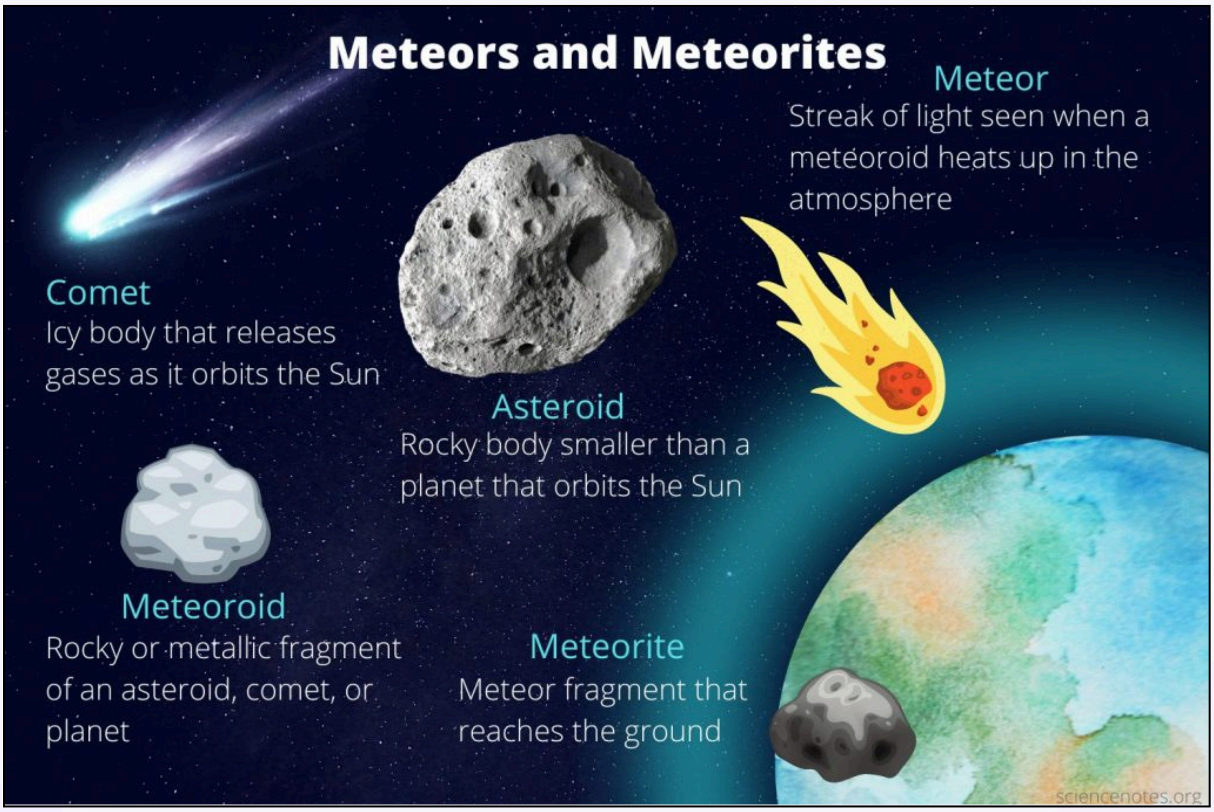
Comets

A comet is an **icy** small Solar System body that, when passing close to the Sun, heats up due to the effects of solar radiation and the solar wind upon the nucleus and begins to outgas, displaying a visible atmosphere or coma, and sometimes also a tail.

- Comets have **highly elliptical orbits,** unlike the planets which have near-circular orbits.
- One of the larger comets is the **Halley's Comet**. The orbit of Halley's Comet brings it close to the Earth every **76 years**. It was last visited in **1986**.



Meteoroid, Meteor and Meteorite



- **Asteroids** in space sometimes collide with each other, forming **meteoroids**. Small pieces of an asteroid are called **meteoroids**.
- **Meteors** are formed from these **meteoroids**.
- When a **meteoroid** enters Earth's atmosphere, it appears as a **streak of light** in the sky, called a **meteor**.
- **Meteors** travel at **high speed** and heat up due to **friction** with the atmosphere, causing them to light up as a streak of light.

- **Meteors** usually **evaporate** in the sky before reaching Earth's surface, so the streak of light is visible only for a **short time**.
- **Meteors** are often called **shooting stars**, but they are not stars; they are simply **pieces of rock**.
- When Earth passes through a **comet's tail**, a large number of **meteors** can be seen in the sky, creating a **meteor shower**.